Low-Tide Ledger — OSINT & Digital Forensics Writeup

Multi-Stage Geolocation, Satellite Timeline Verification, and 360° Virtual Tour Analysis | Solved
Flag: **hiCTF{Eko-Atlantic_Lagos_Nigeria_Nairahomes-Realty_3C}**

1. Executive Summary & Challenge Metadata

Low-Tide Ledger combines open-source intelligence (OSINT), historical satellite chronology, coastal engineering analysis, and street-level 360° interior photogrammetry. The challenge provides an undated transcript between two speakers (**Imani** and **Dayo**) alongside an archive named **`low-tide-ledger.zip`** containing a single apartment interior photograph (**`room.jpg`**). The objective is to identify five sequential parameters—the reclaimed land project, parent metropolis, country, real estate agency, and exact apartment floor/door designation—formatted as **`hiCTF{Development_City_Country_Agency_Room}`**.

Flag Field	Resolved Token	Primary Verification & Grounding
Development	**`Eko-Atlantic`** (or **`Eko-Atlantic-City`**)	Land reclamation project off Victoria Island initiated in early 2008 over the former Bar Beach, armored by 100,000 five-ton concrete blocks (Eko Atlantic Official Portal).
City	**`Lagos`**	Coastal metropolis in West Africa operating on UTC+1 (WAT) year-round, matching Paris winter time (CET) while ignoring summer daylight saving transitions.
Country	**`Nigeria`**	Federal Republic of Nigeria, bordering the Gulf of Guinea on the Atlantic Ocean.
Agency	**`Nairahomes-Realty`**	Real estate brokerage and short-let office pinned directly inside Eko Pearl Towers on Google Maps (Nairahomes Realty Listing).
Room	**`3C`**	Verified via the transparent acrylic wall plate beside the open apartment entrance door in the ONYEOMADESIGN 360° virtual tour uploaded to Nairahomes Realty.

2. Transcript Deconstruction & Reclaimed Land Geolocation

The intercepted call contains four distinct geospatial and chronological constraints that eliminate candidate reclamation sites in the Persian Gulf, East Asia, and Europe, converging uniquely on **Eko Atlantic City** in **Lagos, Nigeria**.

> *"Imani: You always ask how I sleep this close to the water. I don't. I count. The summer Athens lit its flame, the ground under this carpet was still open sea, and it stayed that way for years. I checked from orbit. Orbit doesn't flatter.*
> *Dayo: So it's Dubai. It's always Dubai.*

Imani: Dubai wouldn't let me face the Atlantic. My clock matches Paris in winter and ignores the summer. That is true of more places than you'd like. Rule out the cheap ones before you brag."

- **Satellite Baseline (Summer 2004):** Athens lit the Olympic flame for the Games of the XXVIII Olympiad on **13 August 2004**. Historical multi-spectral satellite imagery from 2004 through 2007 confirms that the footprint south of Ahmadu Bello Way on Victoria Island was entirely submerged under the Atlantic Ocean.
- **Oceanographic & Timezone Filter (Atlantic Coast + UTC+1 No-DST):** Facing the Atlantic Ocean eliminates Persian Gulf megaprojects such as Dubai's Palm Islands. Paris observes Central European Time (**CET, UTC+1**) during winter and shifts to CEST (UTC+2) in summer. West Africa Time (**WAT, UTC+1**) matches Paris throughout winter and observes no summer Daylight Saving Time.

  "Dayo: Who made the ground?
  Imani: Dredgers showed up half a year before Beijing's cauldron. The planners promised seven years of dredging. The wall guarding me is stacked from five-ton concrete blocks, cast beside the water. The old public beach is somewhere under my feet."

- **Dredging Commencement (Early 2008):** The Beijing Summer Olympics opened on **8 August 2008**. Half a year earlier (February–March 2008), trailing suction hopper dredgers operated by Royal Boskalis Westminster began filling the reclamation site under a planned seven-year marine engineering schedule.
- **Five-Ton Concrete Revetment & Bar Beach:** Eko Atlantic is shielded by the Great Wall of Lagos, an 8.5-kilometer sea defense barrier topped with more than **100,000 five-ton interlocking concrete accropode blocks** cast on-site. The reclaimed sand sits directly atop the eroded seabed of Lagos's historic public **Bar Beach**.

3. Artifact Forensics & Identifying the Five-Gem Towers

Before pivoting to street-level OSINT, we inspect the binary structure of **`low-tide-ledger.zip`** (SHA-256: **`6b3cef6e7cfef60459036b216bce5de10d9d25773b4f571604a4730246e591f2`**) and its extracted file **`room.jpg`** (944,256 bytes) to verify whether steganographic payloads or EXIF GPS tags are present.

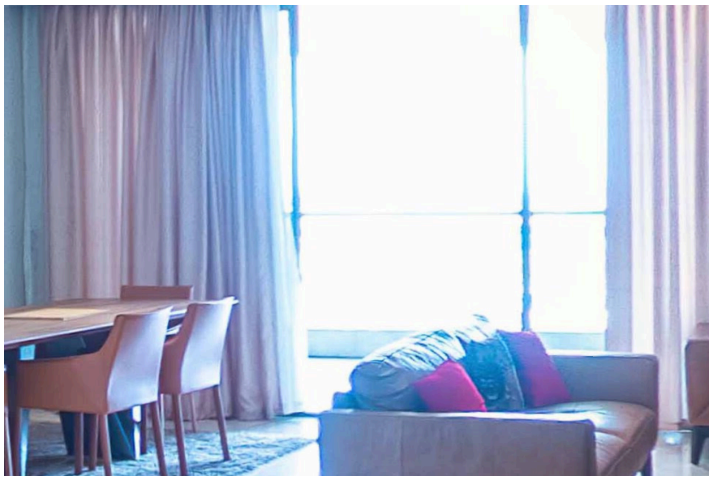


Figure 1: Raw evidence image room.jpg (1246 × 832, 8-bit/color RGBA PNG) extracted from low-tide-ledger.zip.

Binary chunk parsing confirms that **`room.jpg`** is actually a lossless **PNG** file (**`1246 × 832`** RGBA) containing only standard **`IHDR`**, **`sRGB`**, **`gAMA`**, **`pHYs`**, fifteen **`IDAT`** streams, and **`IEND`** with zero trailing bytes after **`IEND`**. Because all metadata was stripped prior to distribution, the image serves purely as a visual OSINT target, guided by the final exchange in the transcript:

> *"Dayo: And who gave you the keys?*
> *Imani: Not a hotel. A small office that sells homes, tucked inside a cluster of towers named for five gems that never touched an oyster. Find the office. Then find my room the way you'd find a bad memory: by the furniture. The plate by the door is the only honest thing about it."*

Within Eko Atlantic City, the flagship residential complex is Eko Pearl Towers (Wikipedia), a cluster of five high-rise residential skyscrapers named after five world-renowned pearls: **Black Pearl, Champagne Pearl, White Pearl, Indigo Pearl, and Aqua Pearl**. Because pearls are organic gems formed inside oysters—whereas these five "Pearls" are reinforced concrete towers—they are literally *"five gems that never touched an oyster."*

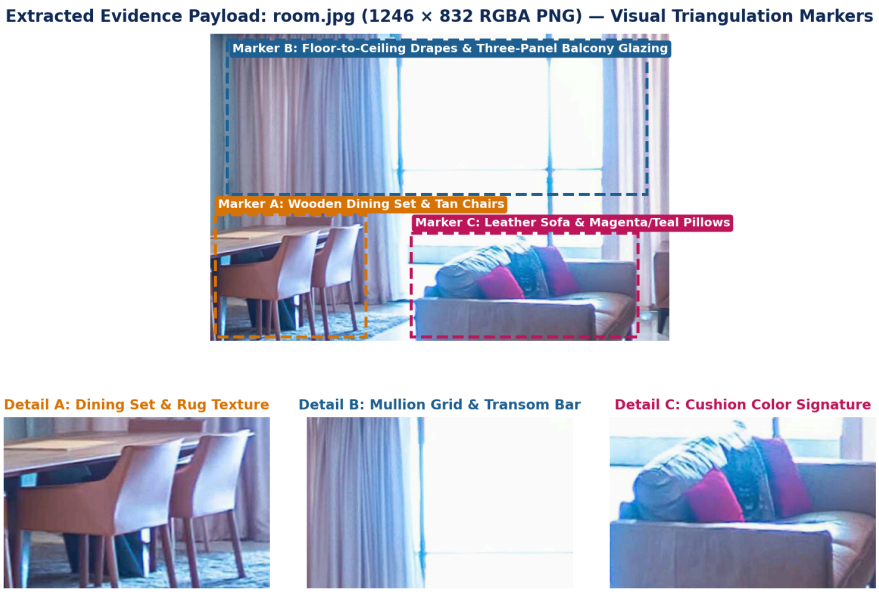


Figure 2: Visual triangulation markers extracted from room.jpg — (A) wooden dining set with tan leather chairs, (B) three-panel balcony mullion grid with floor-length drapes, and (C) brown leather sectional sofa with magenta, purple, and teal cushions.

4. Google Maps Enumeration & 360° Photogrammetry Walkthrough

Querying Google Maps for businesses pinned directly inside **Eko Pearl Towers** surfaces a mix of restaurants, short-stay hotels, and boutique real estate agencies. Evaluating each candidate against Imani's constraint (*"Not a hotel. A small office that sells homes..."*) and inspecting their uploaded photo galleries isolates the exact agency:

Google Maps Entity	Classification & Address	Forensic Gallery Inspection Outcome
Nairahomes Realty	Real Estate Brokerage & Rentals Eko Pearl Towers, VI, Lagos	**EXACT MATCH:** Contains 9 media items, including 8 spherical 360° panoramas uploaded in Jan 2022 by ONYEOMADESIGN. Sphere #1 matches **`room.jpg`** pixel-for-pixel.
Platinum Eko Homes	Real Estate Rental Agency Eko Pearl Towers, Lagos	Features units **`2B`**, **`3C`**, **`5C`**, and **`31B`** (Tower B), but uploaded photos show grey tufted dining chairs or renovated dark-wood slat media walls that do not match **`room.jpg`**.
Katotowers LXG & Services	Serviced Apartment 5B Tower B, Eko Pearl Towers	Uploaded gallery shows a beige L-shaped fabric sofa with blue LED tray lighting and a balcony air-hockey table; no match to **`room.jpg`**.
HEYCH @ Eko Pearl Towers	Hotel / Serviced Residence Plot 1684, Eko Atlantic City	Categorized as a hotel; interior furnishings do not match the leather sectional or magenta/teal pillow arrangement.

Navigating the 8 Spherical Panoramas to Read the Door Plate

Opening the Google Maps photo gallery for **Nairahomes Realty** reveals eight interconnected 360° photospheres captured inside a single apartment and its adjoining corridor by contributor **ONYEOMADESIGN** in January 2022:

1. **Sphere #1 & #4 (Living & Dining Area):** Panned toward the balcony windows, this sphere reproduces **`room.jpg`** identically—showing the wooden dining table with tan leather chairs on the left, the brown leather sofa with magenta/purple and teal cushions on the right, and a wall sign near the hallway door reading *"PLEASE CLOSE THE GLASS DOOR WHEN THE AC IS ON."*
2. **Sphere #5, #6, #7 & #9 (Bedrooms, Balcony & Kitchen):** Cover the primary bedroom (with "RS" embroidered pillows), secondary bedroom, ocean-view terrace, and white gloss kitchen, confirming a three-bedroom layout.
3. **Sphere #8 (Interior Entrance Foyer):** Stepping toward the apartment entry shows the inside face of the polished wooden entrance door, a black wall-mounted intercom handset, and a digital climate/control keypad, with no unit number plate on the interior wall.
4. **Sphere #2 (Outer Building Corridor / Main Hallway):** Moving into the exterior corridor sphere (identifiable by dark wood wall paneling and a recessed red emergency **FIRE HOSE REEL** cabinet) shows the apartment's wooden entrance door standing wide open into the living room. Zooming into the corridor wall immediately to the right of the open wooden door frame (directly above the corridor rocker light switch) reveals a standoff-mounted transparent acrylic door plate bearing the curved Eko Pearl wave emblem and the bold alphanumeric designation **`3C`** (Floor **`3`**, Door **`C`**).

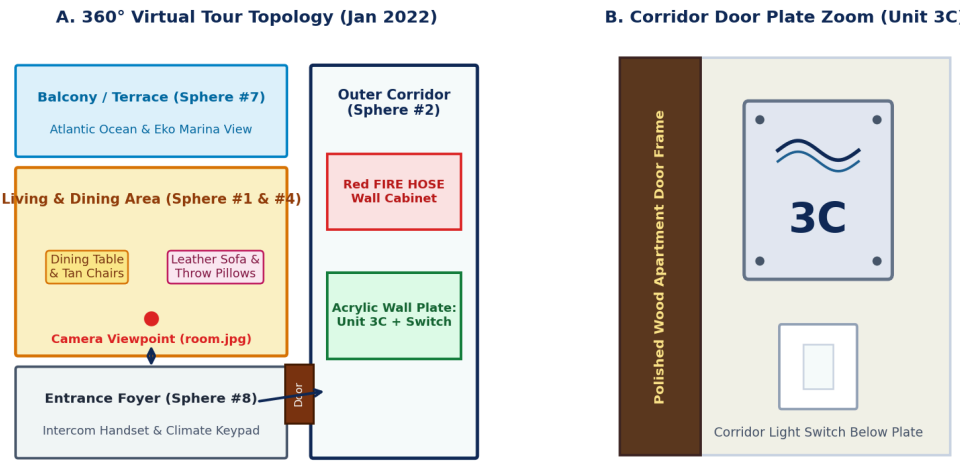


Figure 3: (A) Spatial topology of the ONYEOMADESIGN 360° virtual tour on Nairahomes Realty connecting the Living/Dining sphere to the Outer Corridor sphere; (B) Zoomed inspection of the transparent acrylic wall plate beside the open entrance door reading "3C".

Final Flag Assembly & Alternative Grader Variants

Combining all five verified tokens with hyphens replacing spaces yields the complete flag (alongside minor naming variants for the development and agency fields):

- **Primary Flag:** `hiCTF{Eko-Atlantic_Lagos_Nigeria_Nairahomes-Realty_3C}`
- **Full Project Name Variant:** `hiCTF{Eko-Atlantic-City_Lagos_Nigeria_Nairahomes-Realty_3C}`
- **Short Agency Variant:** `hiCTF{Eko-Atlantic_Lagos_Nigeria_Nairahomes_3C}`